# 1 Worked example: estimability in a two-receiver network

This section makes the rank deficiency and minimum-constraint datum of the minimum-constraint datum section of the Network-mode document concrete on the smallest network that still shows the structure: two receivers $\mathcal{R} = \{A, B\}$ and four satellites $i \in \{1, 2, 3, 4\}$. We strip everything that is not essential to the estimability argument — positions are known so the geometric range is removed, a single frequency is used, the troposphere and ionosphere are absent, and the data are noise-free — so that every number below is *exact* and the gauge freedoms are visible by inspection. The point is not the size of the system but its null space.

## 1.1 Code only: clocks and one gauge freedom

With geometry removed, the single-frequency pseudorange of the observation-equations section collapses to a difference of two clock lumps,

$$P_r^i \;=\; a_r \;-\; g^i, \qquad a_r := c\,\mathrm{d}t_r + B_r, \quad g^i := c\,\mathrm{d}t^i + B^i, \tag{1}$$

where $a_r$ is the receiver clock and $g^i$ the satellite clock. On a single frequency the receiver and satellite *code biases* $B_r, B^i$ are inseparable from the respective clocks, so each "clock" here is already a clock + code-bias lump — the IF-vs-DCB split of the satellite phase-bias section of the Network-mode document degenerates to nothing with one band. There are $|\mathcal{R}| + |\{i\}| = 2 + 4 = 6$ unknowns and $2 \times 4 = 8$ observations.

Take as ground truth (metres)

$$a_A = 5, \quad a_B = 3, \qquad g^1 = 2, \quad g^2 = -1, \quad g^3 = 4, \quad g^4 = 0.5,$$

which through Equation (1) produces the eight pseudoranges

| $P_r^i$ | $i=1$ | $i=2$ | $i=3$ | $i=4$ |
|---|---|---|---|---|
| $r=A$ | 3 | 6 | 1 | 4.5 |
| $r=B$ | 1 | 4 | −1 | 2.5 |

**The deficiency, seen two ways.** *(i) Invariance.* Add any constant $\kappa$ to *all six* unknowns, $a_r \mapsto a_r + \kappa$ and $g^i \mapsto g^i + \kappa$; the difference $a_r - g^i$, and therefore every observation, is unchanged. The truth above and the shifted set $\{15, 13, 12, 9, 14, 10.5\}$ fit the data identically. Absolute clocks are unknowable; only differences are.

*(ii) Singular normals.* Ordering the unknowns $\boldsymbol{x} = [\,a_A, a_B, g^1, g^2, g^3, g^4\,]^{\mathrm{T}}$ and using unit weights, the normal matrix $\boldsymbol{N} = \boldsymbol{A}^{\mathrm{T}}\boldsymbol{A}$ is

$$\boldsymbol{N} = \begin{bmatrix} 4 & 0 & -1 & -1 & -1 & -1 \\ 0 & 4 & -1 & -1 & -1 & -1 \\ -1 & -1 & 2 & 0 & 0 & 0 \\ -1 & -1 & 0 & 2 & 0 & 0 \\ -1 & -1 & 0 & 0 & 2 & 0 \\ -1 & -1 & 0 & 0 & 0 & 2 \end{bmatrix}, \qquad \boldsymbol{N}\,\mathbf{1} = \mathbf{0}, \tag{2}$$

which is $6 \times 6$ but has rank 5: the all-ones vector $\mathbf{1} = [1, 1, 1, 1, 1, 1]^{\mathrm{T}}$ is an exact null vector (every row sums to zero), and it is precisely the common-shift direction of view (i). The system $\boldsymbol{N}\boldsymbol{x} = \boldsymbol{u}$ cannot be solved without one added constraint — a *datum*.

**Two datum choices, same fit.** Pinning one clock removes the gauge. Two natural choices give *different* parameter values that fit the eight observations *equally and exactly*:

| datum (S-basis) | $a_A$ | $a_B$ | $g^1$ | $g^2$ | $g^3$ | $g^4$ |
|---|---|---|---|---|---|---|
| $a_A \equiv 0$ (reference receiver) | 0 | −2 | −3 | −6 | −1 | −4.5 |
| $g^1 \equiv 0$ (reference satellite) | 3 | 1 | 0 | −3 | 2 | −1.5 |

Under $a_A \equiv 0$, every satellite-clock estimate comes out as $\hat{g}^i = g^i - a_A^{\text{true}}$: the reference receiver's clock is *absorbed* into all of them. The estimated "satellite clock" is not the physical clock; it is the physical clock plus the datum we chose — the *lumping* that the user must add back. This is exactly why a producer must *declare* its datum (the closed-loop section of the Network-mode document).

**What is invariant.** The estimable quantities — combinations annihilated by the null vector $\mathbf{1}$, i.e. differences — are identical in both bases:

| estimable quantity | $a_A \equiv 0$ | $g^1 \equiv 0$ | (truth) |
|---|---|---|---|
| $a_A - a_B$ | 2 | 2 | 2 |
| $g^2 - g^1$ | −3 | −3 | −3 |
| $g^3 - g^1$ | 2 | 2 | 2 |
| $a_B - g^4$ | 2.5 | 2.5 | $(= P_B^4)$ |

Only the absolute labels are convention; the differences are physics. This is the S-system of [1] in six parameters.

*Remark* 1 (Why the producer cannot difference). Form the double difference of this code-only model, $\nabla\Delta P_{AB}^{i,1} = (P_A^1 - P_A^i) - (P_B^1 - P_B^i)$. The between-satellite step gives $g^i - g^1$ for *both* receivers, so the between-receiver step yields exactly $\mathbf{0}$. Double differencing annihilates the entire model, because DD is built to delete clocks and here clocks are all there is (the differencing section). A DD user would estimate nothing; the network producer instead *keeps* the clocks and pins one datum. That degeneracy is the producer's reason for existence in miniature.

### 1.2 Adding carrier phase: the satellite phase bias needs the network

Now add the single-frequency carrier phase. With the same clock lumps it reads

$$\Phi_r^i \;=\; a_r - g^i \;+\; b^i \;+\; N_r^i, \tag{3}$$

where $b^i$ is the satellite phase bias — the producer's target — and $N_r^i$ is the float ambiguity (in metres). The receiver phase bias and the receiver code-minus-phase offset are constant per receiver and have been absorbed into $N_r^i$; we keep $b^i$ explicit because it is what the producer must emit. Extend the truth with

$$b^1 = 0.10, \qquad b^2 = -0.20, \qquad b^3 = 0.05, \qquad b^4 = 0.15,$$
$$N_A^i : 10,\ -5,\ 20,\ 8, \qquad\qquad N_B^i : 12,\ -3,\ 18,\ 9,$$

giving the eight carrier-phase observations

| $\Phi_r^i$ | $i=1$ | $i=2$ | $i=3$ | $i=4$ |
|---|---|---|---|---|
| $r = A$ | 13.10 | 0.80 | 21.05 | 12.65 |
| $r = B$ | 13.10 | 0.80 | 17.05 | 11.65 |

**Counting: one receiver vs. two.** The clocks are fixed by the code, so per satellite the phase contributes $b^i + N_r^i$. With *one* receiver this is a single equation $b^i + N_A^i$ in two unknowns: the satellite phase bias has no separate existence at all — it is inseparable from that one receiver's ambiguity, and not even redundant. With *two* receivers there are two equations,

$$b^i + N_A^i, \qquad b^i + N_B^i, \tag{4}$$

in three unknowns $(b^i, N_A^i, N_B^i)$: rank 2, one deficiency per satellite. The estimable two-dimensional space now contains the *between-receiver single-difference ambiguity* $\nabla N_{AB}^i = N_A^i - N_B^i$ — a quantity that simply does not exist with one receiver, and is exactly the integer a user resolves (the differencing section). The second receiver is what gives $b^i$ an identity shared across stations.

**Datum and solution.** The joint code + phase system has a 1-dimensional clock gauge (as before) plus one phase-bias↔ambiguity gauge per satellite: $b^i \mapsto b^i + c_i$, $N_r^i \mapsto N_r^i - c_i$ for all $r$. The minimum-constraint datum used here is the clock pin $a_A \equiv 0$ — S-transform-equivalent to the reference-*satellite* clock pin $g^1 \equiv 0$ that the estimator's S-basis actually declares (the minimum-constraint datum section of the Network-mode document: the shared-block gauge pins a satellite clock per constellation, not a receiver clock) — plus the reference-receiver ambiguity pins $N_A^i \equiv 0$ for every $i$ (the local datum on the pivot receiver, folded into station $A$'s rows). Solving gives

| | $i=1$ | $i=2$ | $i=3$ | $i=4$ |
|---|---|---|---|---|
| emitted phase bias $\bar{b}^i = b^i + N_A^i$ | 10.10 | −5.20 | 20.05 | 8.15 |
| $\hat{N}_A^i$ (datum) | 0 | 0 | 0 | 0 |
| $\hat{N}_B^i = N_B^i - N_A^i$ | 2 | 2 | −2 | 1 |

(the clocks are unchanged from Section 1.1). One checks the fit directly, e.g. $\hat{\Phi}_B^3 = a_B - g^3 + \bar{b}^3 + \hat{N}_B^3 = -2 - (-1) + 20.05 + (-2) = 17.05$, matching the observation.

**Reading the result.** Two things are worth dwelling on.

- **The emitted bias is dominated by the lump.** The physical satellite phase bias of satellite 1 is $b^1 = 0.10$ m, but the *emitted* estimable bias is $\bar{b}^1 = 10.10$ m: it has absorbed receiver $A$'s ambiguity (10 m) for that satellite. This is Odijk's $\bar{b}^i = b^i + (\text{pivot-receiver ambiguity})$ (the satellite phase-bias section of the Network-mode document, Eq. 4/9) made arithmetic. A consumer recovers integers only by removing the *same* lumped quantity — producer datum and consumer integer recovery must be one S-system.
- **The network unlocks the integer.** The estimable $\hat{N}_B^i = N_B^i - N_A^i = \nabla N_{AB}^i$ is the between-receiver single-difference ambiguity, the very quantity an RTK user fixes to an integer. It is identically absent in a single-receiver problem; the second station is what brings it into existence. The satellite phase bias becomes a definite, transferable number for the price of pinning one reference ambiguity — and that pin is what it carries.

This four-satellite, two-receiver instance is, up to the geometry and the second frequency it suppresses, exactly the synthetic network the estimator core is validated on: the `RecoversSyntheticTruthUpToSTransform` test (`tests/test_network_estimator.cpp`) builds rows of exactly this shape — at four receivers and five satellites, from a random truth — pins the reference-satellite clock datum $g^1 \equiv 0$ (S-transform-equivalent to the $a_A \equiv 0$ used above) through `NetworkEstimator::solve`, and asserts recovery of the satellite-side parameters *up to the declared S-transform* — i.e. recovery of the lumped $\bar{b}^i$ and the single-difference ambiguities above, not of the unobservable physical $b^i$.

## References

[1] P. J. G. Teunissen. "Generalized inverses, adjustment, the datum problem and S-transformations". In: *Optimization and Design of Geodetic Networks* (1985), pp. 11–55.